

Cryptology - 4

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1) Decrypt the following ciphertext using the Hill cipher.

* Ciphertext = $\begin{matrix} 7 & 11 & 3 & 19 & 16 & 8 & 7 & 9 & 3 & 23 & 22 & 16 & 2 & 12 & 0 & 6 \\ H & L & D & T & Q & I & H & J & D & X & W & Q & C & M & A & G \end{matrix}$

Key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ Find Plain-text-

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \quad : \quad 7 \times 3 - 2 \times 5 = 21 - 10 = 11$$

$$11 \bmod 26 = ?$$

$$26 = 11(2) + 4$$

$$26 + 11(-2) = 4 \rightarrow \textcircled{1}$$

$$11 = 4(2) + 3$$

$$11 + 4(-2) = 3 \rightarrow \textcircled{2}$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$4 + 11(-1) + 4(2) = 1$$

$$4(3) + 11(-1) = 1$$

$$26(3) + 11(-6) + 11(-1) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 1$$

$$11^{-1} \bmod 26 = 26 - 7 = \underline{\underline{19}}$$

$$K^{-1} = 19 \begin{bmatrix} 3 & -2 \\ -5 & 7 \end{bmatrix} \text{mod } 26.$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{mod } 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \rightarrow \textcircled{1}.$$

$$\text{Plain text} = \text{Ciphertext } [K^{-1}] \text{mod } 26.$$

$$H=7$$

$$L=11$$

$$P.T = K^{-1} C.T \text{mod } 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix}$$

$$P.T = \begin{bmatrix} 3 \times 7 + 14 \times 11 \\ 9 \times 7 + 5 \times 11 \end{bmatrix} = \begin{bmatrix} 21 + 154 \\ 63 + 55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \text{mod } 26$$

$$P.T = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} T \\ O \end{bmatrix} \rightarrow \textcircled{2}$$

$$D=3$$

$$T=19$$

$$P.T = K^{-1} C.T \text{mod } 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \text{mod } 26$$

$$P.T = \begin{bmatrix} 3 \times 3 + 14 \times 19 \\ 9 \times 3 + 5 \times 19 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 275 \\ 122 \end{bmatrix} = \begin{bmatrix} 15 \\ 18 \end{bmatrix}$$

$$P.T = PS$$

$$Q = 16 \quad PT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \text{ mod } 26$$

$$I = 8$$

$$PT = \begin{bmatrix} 3 \times 16 + 14 \times 8 \\ 9 \times 16 + 5 \times 8 \end{bmatrix} = \begin{bmatrix} 48 + 112 \\ 144 + 40 \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \text{ mod } 26$$

$$PT = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{matrix} E \\ C \end{matrix}$$

$$\boxed{PT = EC}$$

$$H = 7$$

$$PT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \text{ mod } 26$$

$$J = 9$$

$$PT = \begin{bmatrix} 3 \times 9 + 14 \times 9 \\ 9 \times 7 + 5 \times 9 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 21 + 126 \\ 63 + 45 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \text{ mod } 26$$

$$PT = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$\boxed{PT = RE}$$

$$D = 3$$

$$PT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \times \begin{bmatrix} 3 \\ 23 \end{bmatrix} \text{ mod } 26$$

$$K = 23$$

$$PT = \begin{bmatrix} 3 \times 3 + 14 \times 23 \\ 9 \times 3 + 5 \times 23 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \text{ mod } 26$$

$$PT = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$\boxed{PT = TM}$$

$$W = 22$$

$$Q = 16$$

$$PT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \times \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 3 \times 22 + 14 \times 16 \\ 9 \times 22 + 5 \times 16 \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix}$$

$$PT = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} E \\ S \end{matrix}$$

$$C=2 \quad PT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \times \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 2 + 14 \times 12 \\ 9 \times 2 + 5 \times 12 \end{bmatrix}$$

$$M=12 \quad PT = \begin{bmatrix} 6 + 168 \\ 18 + 60 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{bmatrix} S \\ A \end{bmatrix}$$

$$PT = SA$$

$$PT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \times \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 0 + 14 \times 6 \\ 9 \times 0 + 5 \times 6 \end{bmatrix}$$

$$A=0$$

$$G=6$$

$$PT = \begin{bmatrix} 0 + 84 \\ 0 + 30 \end{bmatrix} = \begin{bmatrix} 12 \\ 4 \end{bmatrix} = \begin{bmatrix} M \\ E \end{bmatrix}$$

$$PT = GME$$

$\therefore PT =$ Top Secret Message.

2 Consider the Hill cipher Matrix.

$$\begin{bmatrix} 24 & 19 \\ 5 & 4 \end{bmatrix}$$

Ciphertext : RXADY

Find the plain text for the above ciphertext.

$$\Rightarrow \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} \quad [24 \times 14 - 19 \times 5] = 241 \text{ mod } 26$$

$$\text{determinant} = \underline{7}$$

$$7^{-1} = 7 \text{ mod } 26$$

$$26 = 7(3) + 5 \Rightarrow 26 + 7(-3) = 5$$

$$7 = 5(1) + 2 \Rightarrow 7 + 5(-1) = 2$$

$$5 = 2(2) + 1 \Rightarrow 5 + 2(-2) = 1$$

$$5 + (-2)(7 + 5(-1)) = 1$$

$$5 + 7(-2) + 5(1) = 1$$

$$5(2) + 7(-2) = 1$$

$$(26 + 7(-3))2 + 7(-2) = 1$$

$$26(2) + 7(-6) + 7(-2) = 1$$

$$26(2) + 7(-8) = 1$$

$$\boxed{7 \text{ mod } 26 = 15}$$

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

RX AO DY
17 23 0 14 3 24

Decrypt : RX

$$P = K^{-1}(C) \pmod{26}$$

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \pmod{26}$$

$$P = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3 \times 17 + 22 \times 23 \end{bmatrix} = \begin{bmatrix} 57 \\ 559 \end{bmatrix} \pmod{26}$$

$$P = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

Decrypt:

A
O

$$= \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 0 + 14 \\ 0 + 22 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} A \\ O \end{bmatrix} = \begin{bmatrix} 0 \\ 14 \end{bmatrix}$$

Decrypt D
Y = 3 24

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \pmod{26}$$

$$P = \begin{bmatrix} 2 \times 3 + 1 \times 24 \\ 3 \times 3 + 22 \times 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 531 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

∴ Plain text = FLOWER

(2) Confusion v/s Diffusion.

Confusion	Diffusion.
Hides the relationship between key & Ciphertext	Hides the relationship between plaintext & ciphertext.
Makes the key - ciphertext relationship complicated	Spreads plaintext information across many cipher bits.
Mainly achieved using substitution	Mainly achieved by permutation / transposition
changing the key should greatly change the cipher text	Changing one plaintext bit should affect many ciphertext bits

Cipher	Does it change the character/ symbol value	Does it change the character position?	Confusion/ Diffusion	Justification
Shift cipher	Yes	No	Confusion	Each character is shifted by a fixed amount, changing its value but not its position.
Multiplicative cipher	Yes	No	Confusion	Each character's numerical value is multiplied by a key, changing the symbol but keeping its position.
Vigenere cipher	Yes	No	Confusion	It uses multiplication & addition to change the character value while positions remain unchanged.
Transposition cipher	No	Yes	Diffusion	The characters themselves remain the same, but their positions are rearranged.

Relationship between Substitution, Confusion, Transposition & Diffusion.

1] Substitution & Confusion.

→ Substitution replace one character @ symbol with another. Since the character value changes but its position usually remains the same, it makes the relationship between the plaintext, ciphertext, & key difficult to understand.

Eg: HELLO → KHOOR. [Shift cipher]

∴ The letters are changed, but their positions remain the same.

2] Transposition & ~~Confusion~~ Diffusion.

Transposition does not change the actual characters. Instead, it rearranges their positions.

This spreads the information from the plaintext to different positions in the ciphertext, making the original message harder to identify.

PT ⇒ HELLO

Key: 32415

CT ⇒ LE LHO

The same letters are present but their positions are rearranged.

∴ Substitution → changes the symbol → provides Confusion.

Transposition → changes character → provides Diffusion.

Transposition:

Encrypt the message "the arrow is pointing north direction" using.

- simple with key 4
 - keyword with keyword = ATLAS
- double
key 1 = BREAK
key 2 : KEY

Solu :-

Transposition cipher technique in which position of characters or rearranged according to specific rule @ key.

> The character themselves don't change

> Only their order / position change

> if n mainly associated with different

Eg: → HELLO → LHOEL

The letters are same, but positions are changed.

Decryption

$$C = T_K(P)$$

Encryption

$$P = T_K^{-1}(C)$$

The Arrow is pointing north direction

Key: 4

THE ARROW IS POINTING NORTH DIRECTION

T H E A

R R O W

I S P O

I N T I

N G N O

R T H D

I R E C

T I O N

ciphertext:

TRIRNRTHRSNGTRIE
OPINHEDAW DIOACN

② keyword : ATLAS

1 5 3 2 4

1 → 2 → 3 → 4 → 5

A A L S T

A T L A S

1 5 3 2 4

T H E A R

R O W I S

P O I N T

I N G N O

R T H D I

R E C T I

O N

A = TRPIRROW

A = AI NNDT

L E N I G H C

S R S T O I I

T H O O N T E N

Ciphertext -

TRPIRROW AIN NDT E W I G H C R S T O I
H O O N T E N

© Double:

key 1: BREAK

key 2: LOG

B R E A K
2 5 3 1 4

T H E A R
R O W I S
P O I N T
I N G N O
R T H D I
R E C T I
O N

1st transposition Cipher

⇒ A I N D T T R P I R R O E W I G H C R S
T O I I H O O N T E N

key 2

L O G
2 3 1

A I N
D T T
R P I
R R O
E W I
G H C
R S T
O I I
H O O
N T E
N

2. The Cipher text

N T I O I C T I B E A D R R E G
R O H N N T T P R W H S I O T

Decryption:

1) Ciphertext : T I D I E M R N M E R E O G O
S A N O W R S T L P E E E D O S S S N U A H T U D
B I E N P G O D A E P E I X D E L N S X .

$$\text{Key} = 10, \div \frac{60}{10} : 6 \text{ rows}$$

	1	2	3	4	5	6	7	8	9	10
1	T	R	O	O	P	S	T	N	E	E
2	I	N	G	W	E	S	U	P	P	L
3	D	M	O	R	E	N	D	G	E	N
4	I	E	S	S	E	O	B	O	I	S
5	E	R	A	T	D	A	I	D	X	X
6	M	E	N	L	O	H	E	A	P	

TROOPS HEADIN G WEST NEED MORE
SUPPLIES SEND G GENERAL DUBOIS
MEN TO AID.

2) Cipher : I L E D T N H P I R S O I I T
 M A X S A X E R X T A N D N I S N E O F X

Key : S E C R E T
 5 2 1 4 3 6

$\frac{36 \text{ letters}}{+ 6} : 6 \text{ rows}$

	5	2	1	4	3	6
1	S	E	C	R	E	T
2	T	H	I	S	I	N
3	A	P	L	A	I	F
4	N	I	E	X	T	O
5	O	R	D	E	M	T
6	N	S	T	R	A	X
7	I	O	N	X	X	

THIS IS A PLAIN TEXT FOR DEMONSTRATION.

3) Cipher text

V E Y E A X A R A A T X H G D

Keyword PLAIN

Total letters = $15 \div 3 = 3 \text{ rows}$

	P	L	A	I	N
	5	3	1	2	4
1	H	A	V	E	A
2	G	R	E	A	T
3	D	A	Y	X	X

PLAINTEXT: HAVE A GREAT DAY

④ Find inverse permutation of $P = [4, 3, 1, 5, 2]$

pos.	1	2	3	4	5
Perm key	4	3	1	5	2

$$P^{-1} = (3, 5, 2, 1, 4)$$

② $P = 7, 4, 1, 3, 5, 2, 6$

Pos	1	2	3	4	5	6	7
Permutation table	7	4	1	3	5	2	6
P^{-1}	3	6	4	2	5	7	1

③ $P = (4, 1, 3, 2)$

Pos	1	2	3	4
P	4	1	3	2

$$P^{-1} = 2, 4, 3, 1$$

④ $P = [3, 5, 1, 4, 2]$

Pos	1	2	3	4	5
P	3	5	1	4	2

$$P^{-1} = 3, 5, 1, 4, 2$$